SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems
COURSE CODE: CSA0502

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Concept Mapping (Medium)
Assessment Tool Weightage: 35%

Dr Ayyapan G

QUESTIONS Total Marks: 50

Q.No	Question	Bloom's Level	Marks
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand	**5**
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand	**5**
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand	**5**
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand	**5**
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand	**5**
6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand	**5**
7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand	**5**

8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand	**5**
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand	**5**
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.	BL1 – Remember	**5**

Code of Conduct:

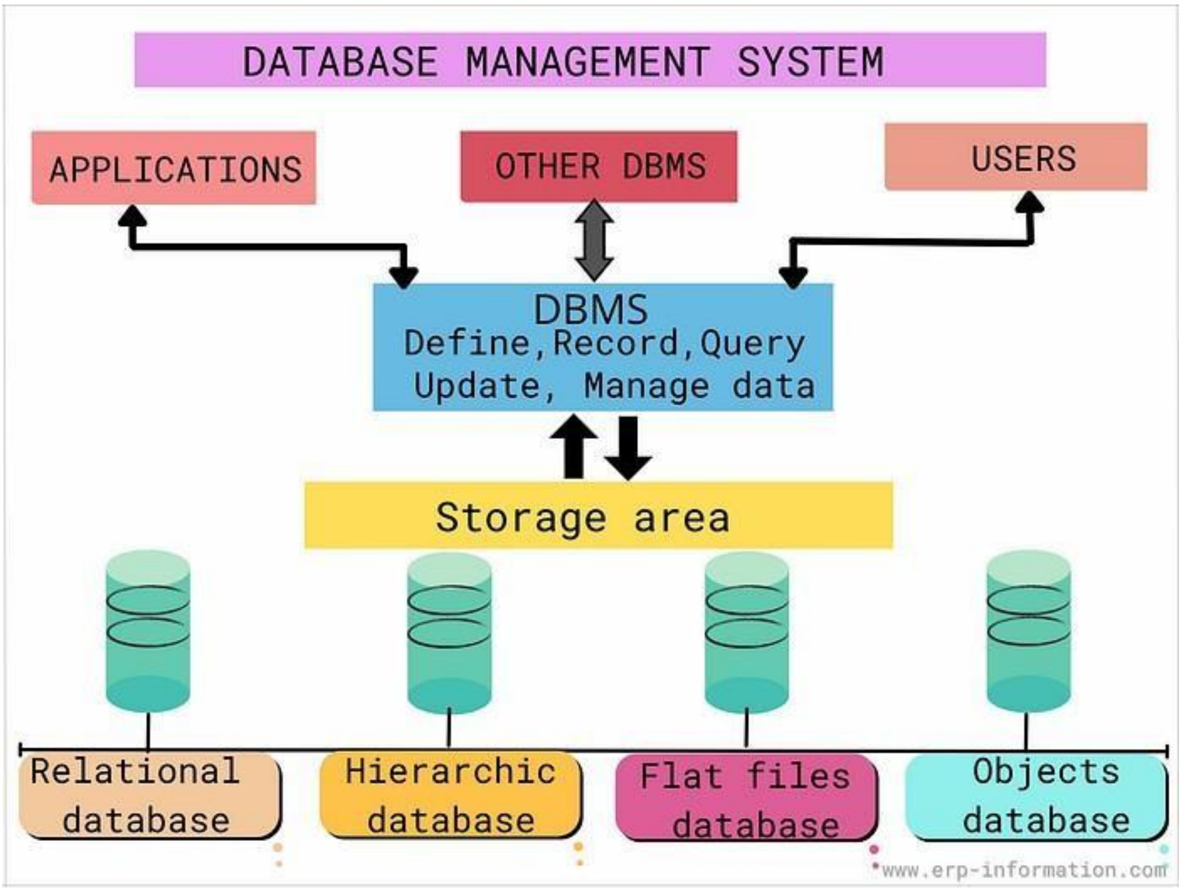
I ***CHINTHALA SRAVANTHI / 192525463*** *Certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

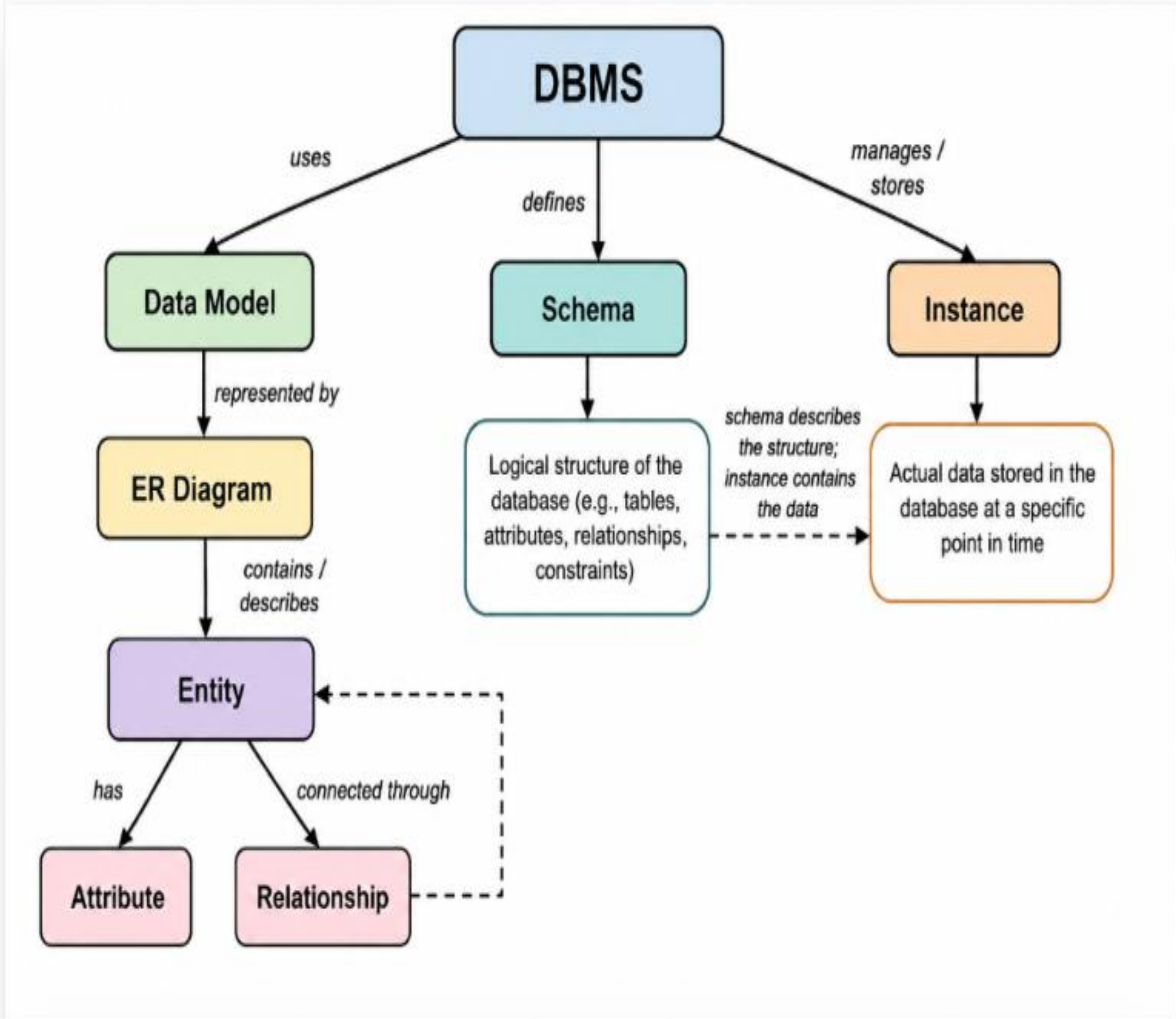
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence

Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

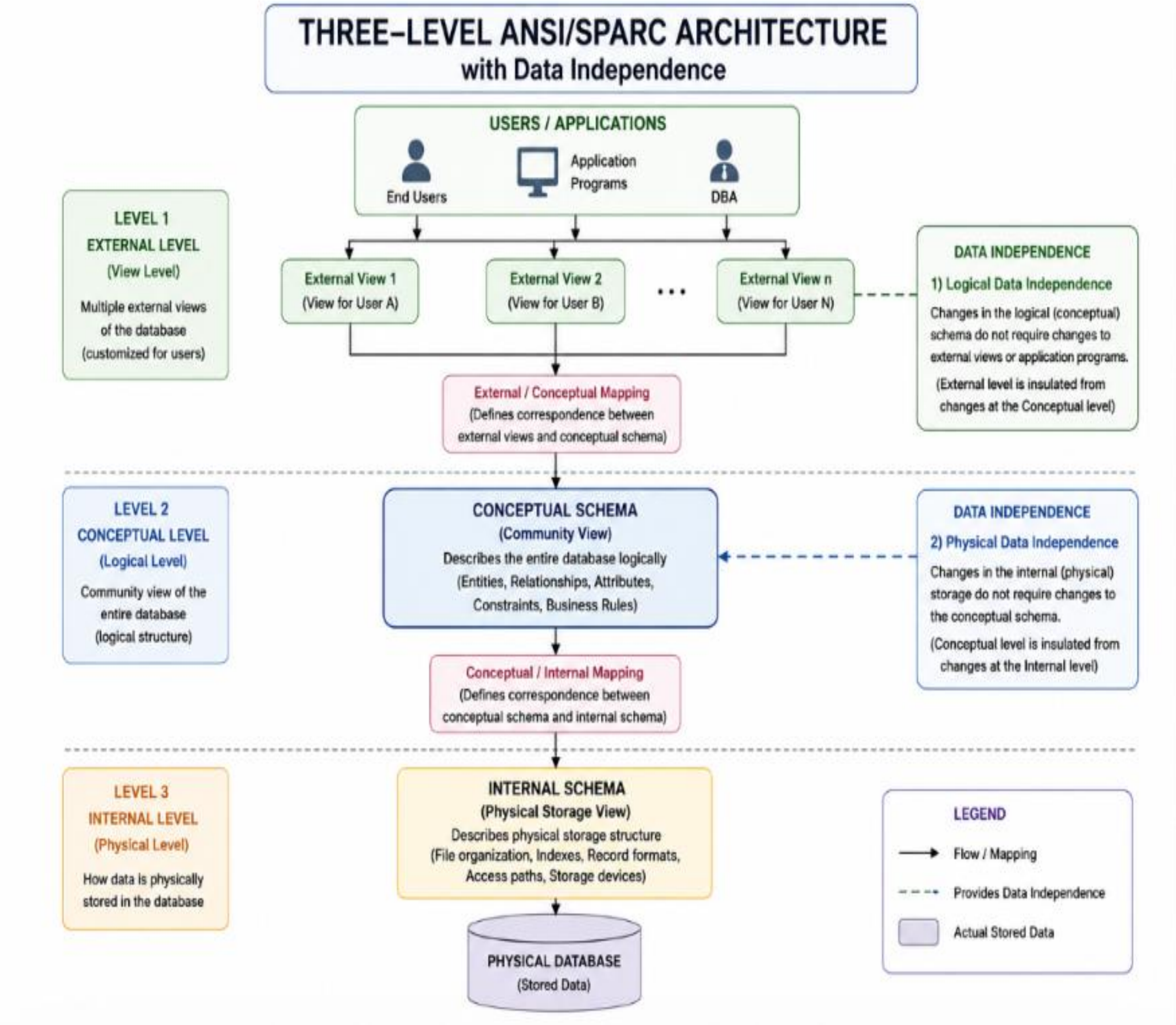


1.

Answers


DBMS
uses
defines
manages / stores
Data Model
Schema
Instance
represented by
ER Diagram
Logical structure of the database (e.g., tables, attributes, relationships, constraints)
schema describes the structure; instance contains the data
Actual data stored in the database at a specific point in time
contains / describes
Entity
has
connected through
Attribute
Relationship

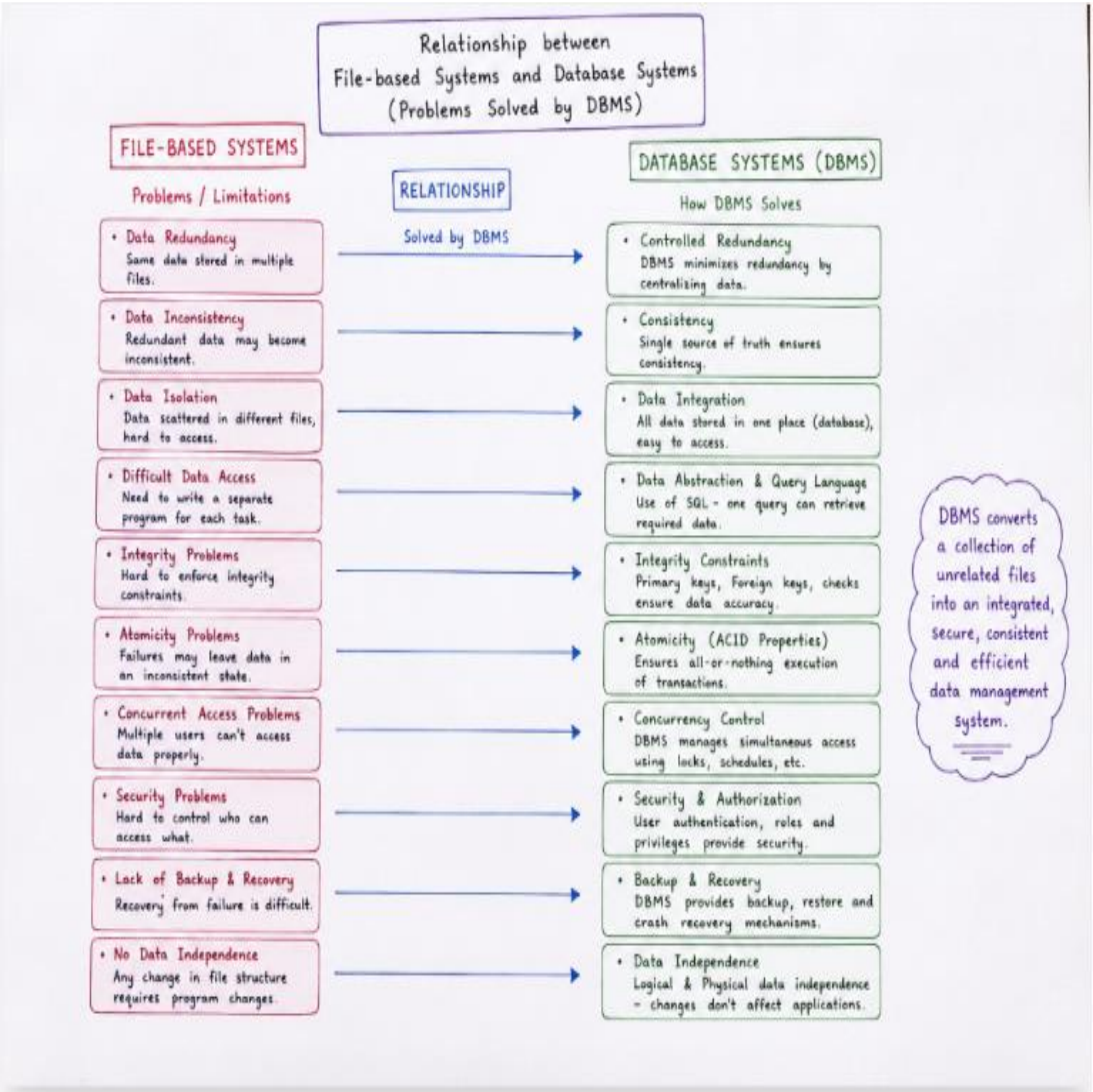
Question 2:



- **Logical Data Independence** is achieved at the boundary between the **External Level** (User Views) and the **Conceptual Level** (Logical Schema).

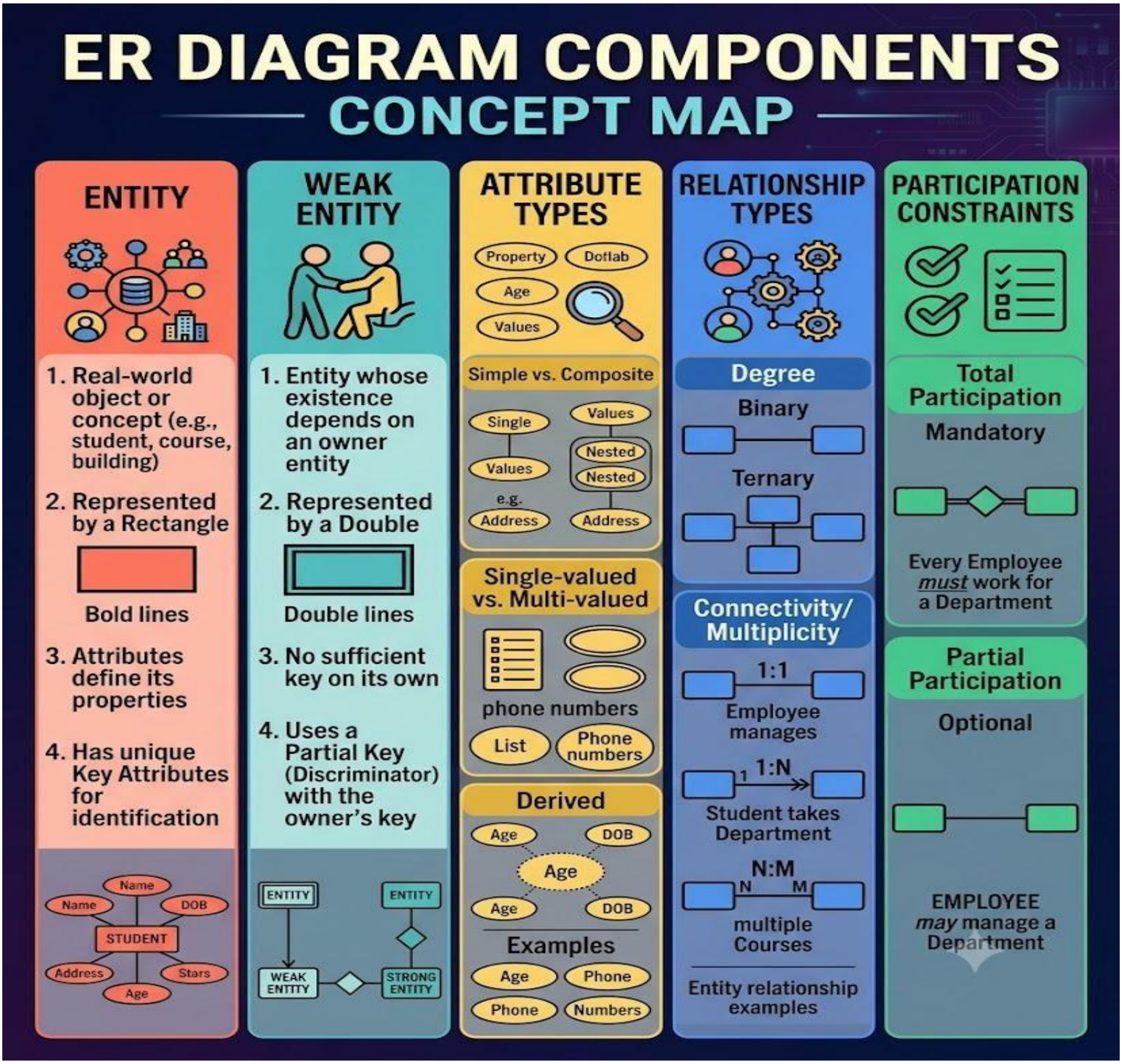
- **Physical Data Independence** is achieved between the **Conceptual Level** and the **Internal Level** (Physical Storage).

Question 3:

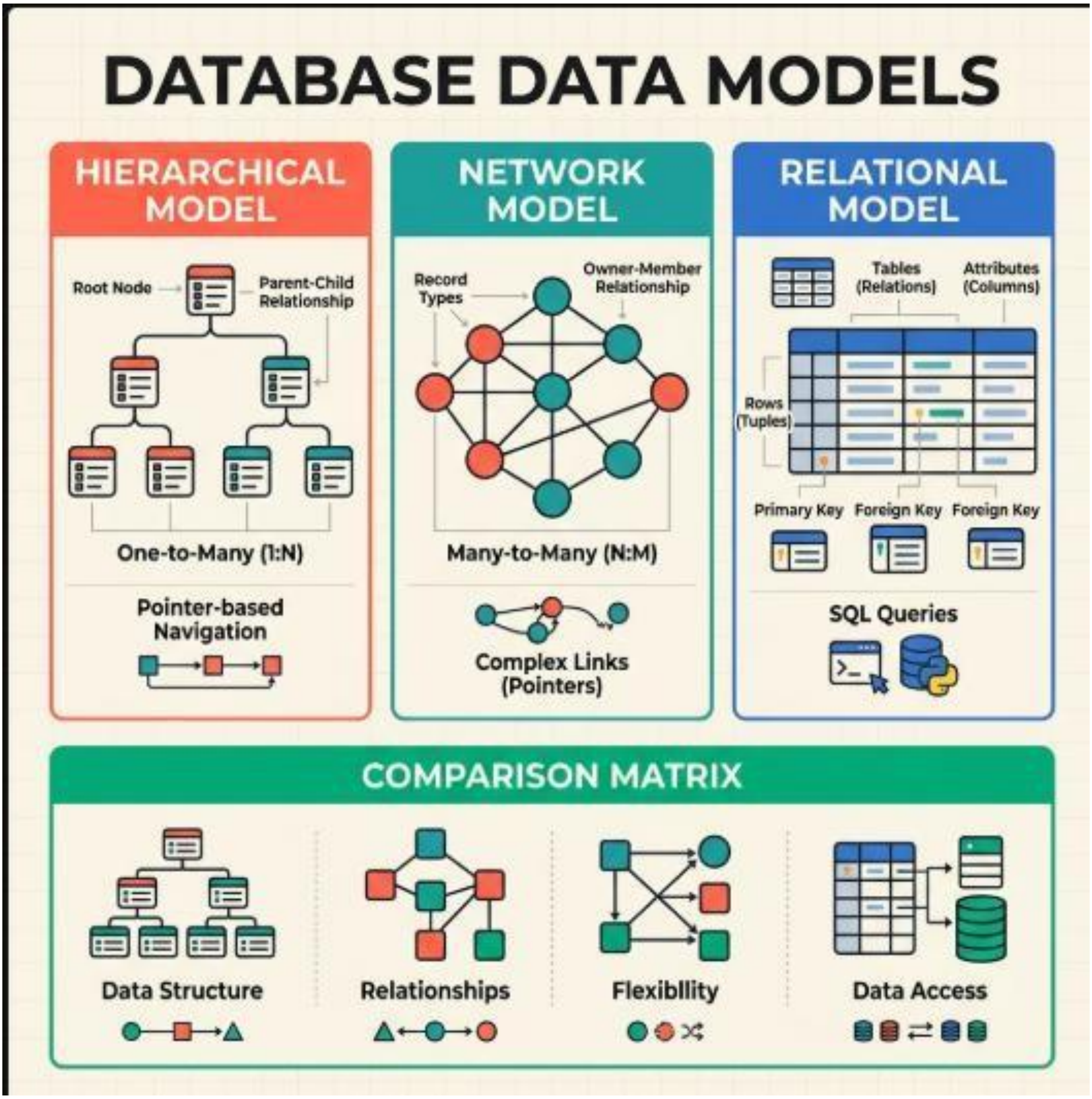


Relationship between
File-based Systems and Database Systems
(Problems Solved by DBMS)
FILE-BASED SYSTEMS
Problems / Limitations
RELATIONSHIP
Solved by DBMS
DATABASE SYSTEMS (DBMS)
How DBMS Solves
• Data Redundancy
Same data stored in multiple files.
• Data Inconsistency
Redundant data may become inconsistent.
• Data Isolation
Data scattered in different files, hard to access.
• Difficult Data Access
Need to write a separate program for each task.
• Integrity Problems
Hard to enforce integrity constraints.
• Atomicity Problems
Failures may leave data in an inconsistent state.
• Concurrent Access Problems
Multiple users can't access data properly.
• Security Problems
Hard to control who can access what.
• Lack of Backup & Recovery
Recovery from failure is difficult.
• No Data Independence
Any change in file structure requires program changes.
• Controlled Redundancy
DBMS minimizes redundancy by centralizing data.
• Consistency
Single source of truth ensures consistency.
• Data Integration
All data stored in one place (database), easy to access.
• Data Abstraction & Query Language
Use of SQL - one query can retrieve required data.
• Integrity Constraints
Primary keys, Foreign keys, checks ensure data accuracy.
• Atomicity (ACID Properties)
Ensures all-or-nothing execution of transactions.
• Concurrency Control
DBMS manages simultaneous access using locks, schedules, etc.
• Security & Authorization
User authentication, roles and privileges provide security.
• Backup & Recovery
DBMS provides backup, restore and crash recovery mechanisms.
• Data Independence
Logical & Physical data independence - changes don't affect applications.
DBMS converts a collection of unrelated files into an integrated, secure, consistent and efficient data management system.

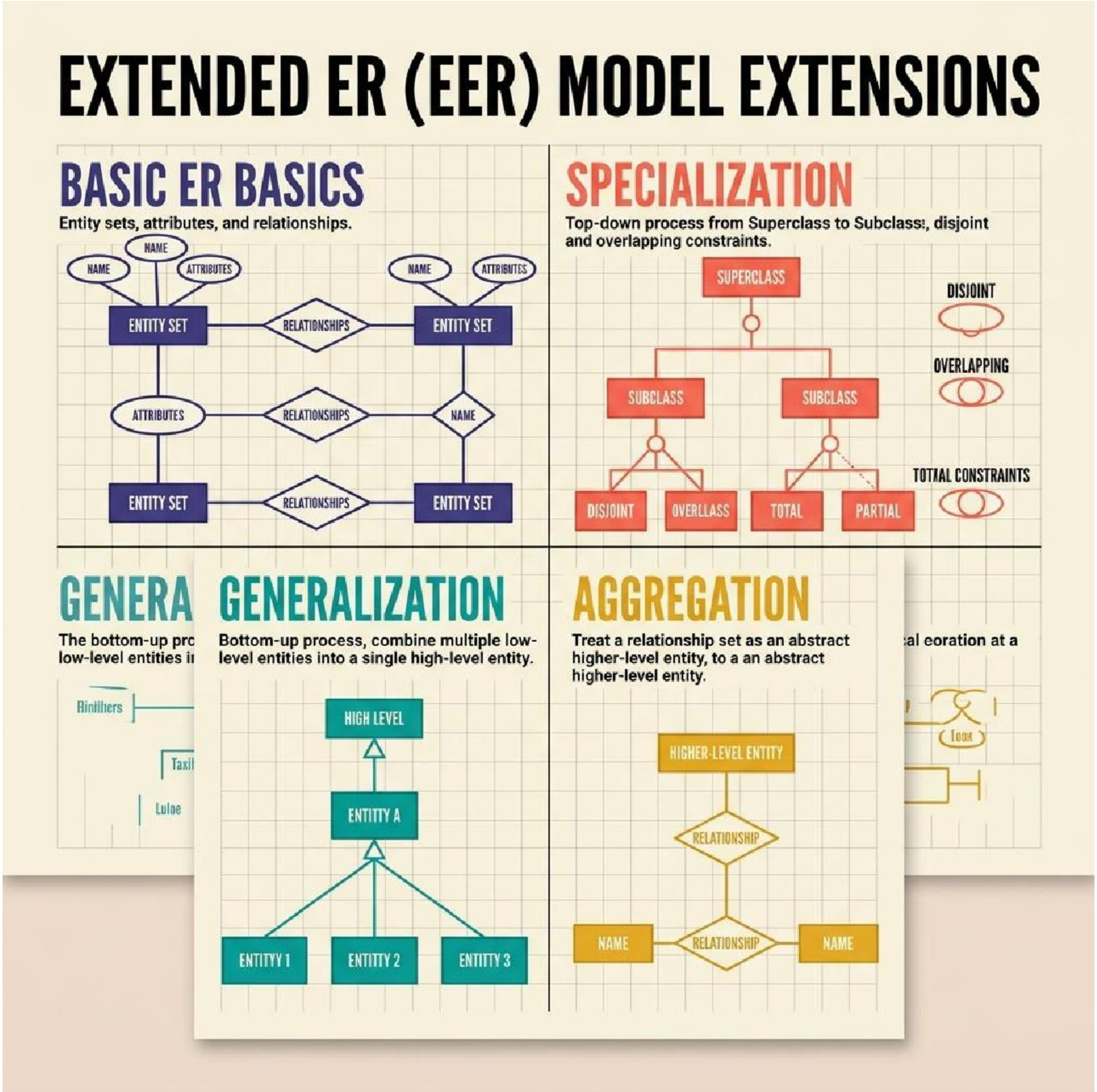
Questions 4:


ER DIAGRAM COMPONENTS
CONCEPT MAP
ENTITY
1. Real-world object or concept (e.g., student, course, building)
2. Represented by a Rectangle
Bold lines
3. Attributes define its properties
4. Has unique Key Attributes for identification
Name
Name
DOB
STUDENT
Address
Stars
Age
WEAK ENTITY
1. Entity whose existence depends on an owner entity
2. Represented by a Double
Double lines
3. No sufficient key on its own
4. Uses a Partial Key (Discriminator) with the owner's key
ENTITY
ENTITY
WEAK ENTITY
STRONG ENTITY
ATTRIBUTE TYPES
Property
Dotlab
Age
Values
Simple vs. Composite
Single
Values
Values
Nested
Nested
e.g.
Address
Address
Single-valued vs. Multi-valued
phone numbers
List
Phone numbers
Derived
Age
DOB
Age
Age
DOB
Examples
Age
Phone
Phone
Numbers
RELATIONSHIP TYPES
Degree
Binary
Ternary
Connectivity/ Multiplicity
1:1
Employee manages
1:N
Student takes Department
N:M
multiple Courses
Entity relationship examples
PARTICIPATION CONSTRAINTS
Total Participation
Mandatory
Every Employee must work for a Department
Partial Participation
Optional
EMPLOYEE may manage a Department

Questions 5:

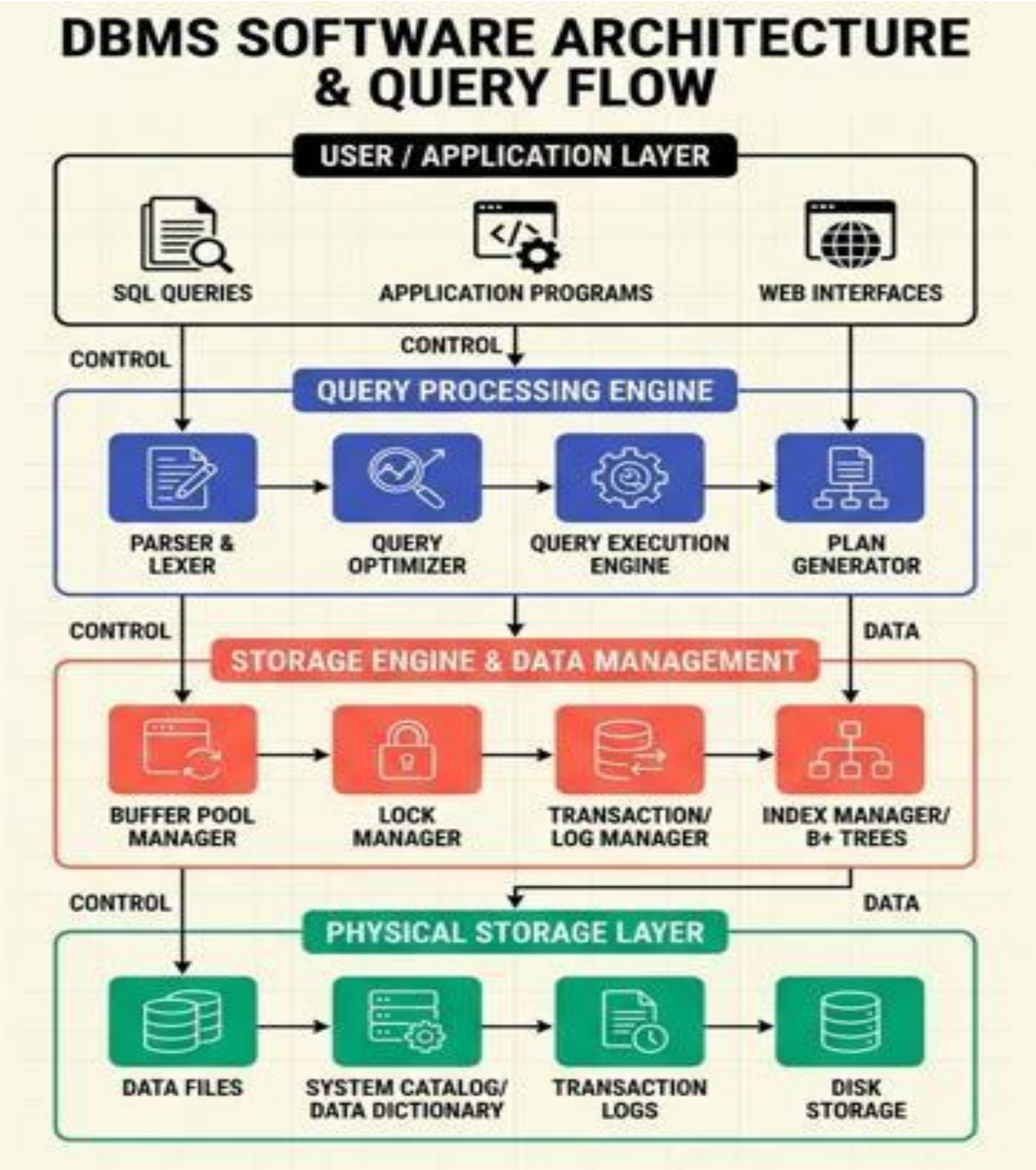

DATABASE DATA MODELS
HIERARCHICAL MODEL
Root Node
Parent-Child Relationship
One-to-Many (1:N)
Pointer-based Navigation
NETWORK MODEL
Record Types
Owner-Member Relationship
Many-to-Many (N:M)
Complex Links (Pointers)
RELATIONAL MODEL
Tables (Relations)
Attributes (Columns)
Rows (Tuples)
Primary Key
Foreign Key
Foreign Key
SQL Queries
COMPARISON MATRIX
Data Structure
Relationships
Flexibllity
Data Access

Question 6:


EXTENDED ER (EER) MODEL EXTENSIONS
BASIC ER BASICS
Entity sets, attributes, and relationships.
NAME
NAME
ATTRIBUTES
NAME
ATTRIBUTES
ENTITY SET
RELATIONSHIPS
ENTITY SET
ATTRIBUTES
RELATIONSHIPS
NAME
ENTITY SET
RELATIONSHIPS
ENTITY SET
SPECIALIZATION
Top-down process from Superclass to Subclass:, disjoint and overlapping constraints.
SUPERCLASS
DISJOINT
OVERLAPPING
SUBCLASS
SUBCLASS
TOTAL CONSTRAINTS
DISJOINT
OVERLLASS
TOTAL
PARTIAL
GENERA
The bottom-up pro
low-level entities i
GENERALIZATION
Bottom-up process, combine multiple low-level entities into a single high-level entity.
HIGH LEVEL
ENTITY A
ENTITYY 1
ENTITY 2
ENTITY 3
AGGREGATION
Treat a relationship set as an abstract higher-level entity, to a an abstract higher-level entity.
al eoration at a
HIGHER-LEVEL ENTITY
RELATIONSHIP
NAME
RELATIONSHIP
NAME

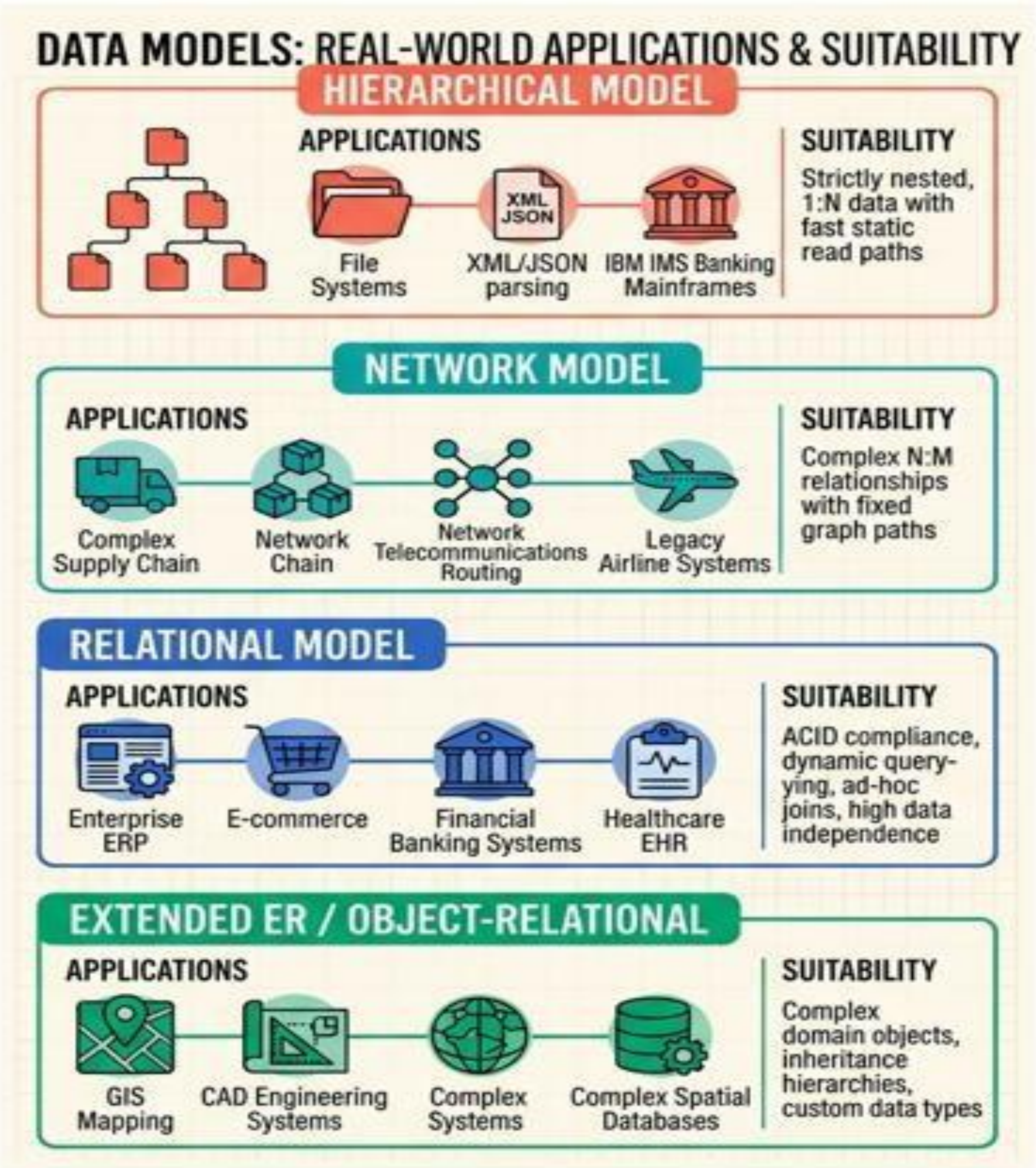
Question 7:

DBMS Software Architecture

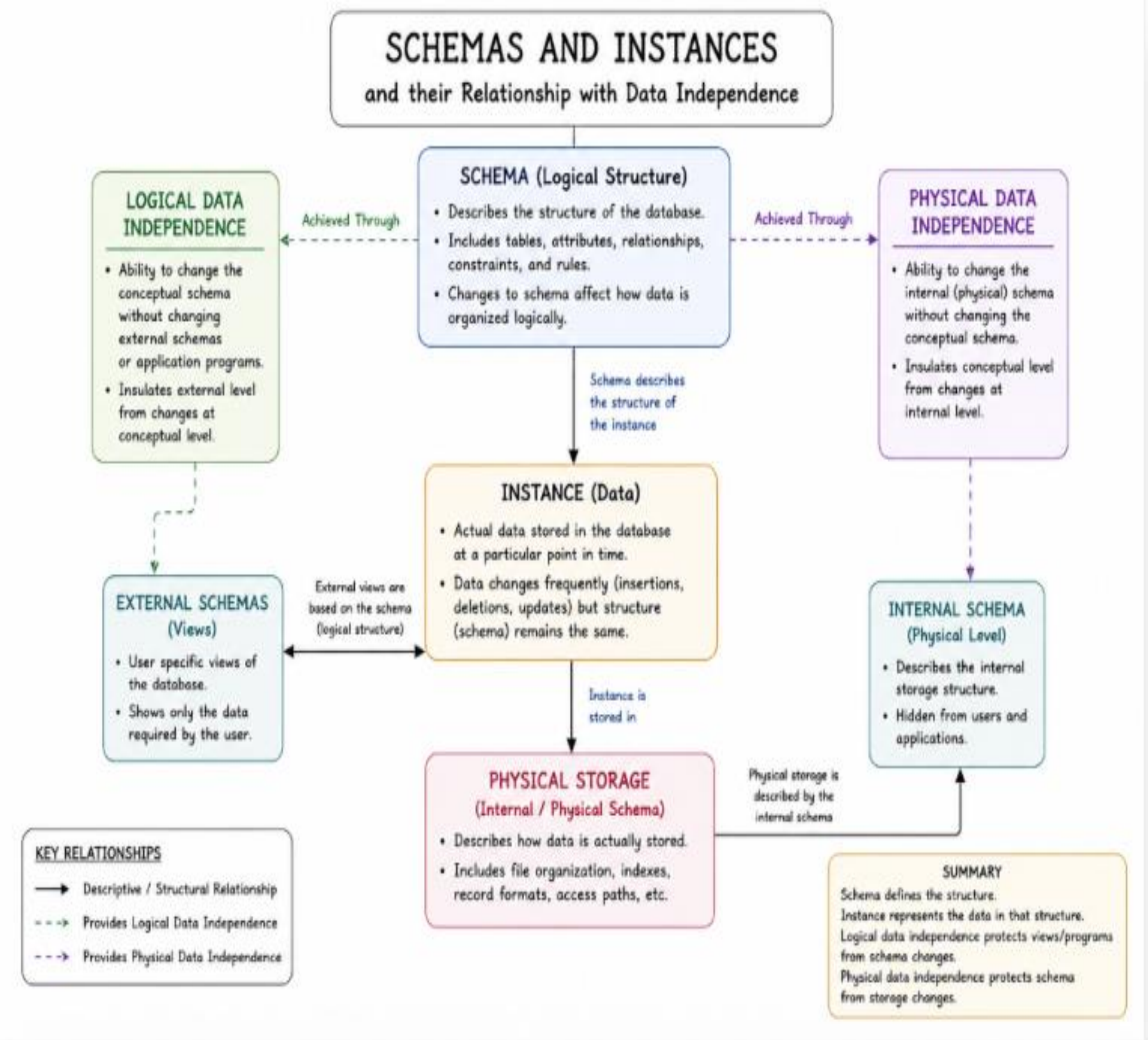


Question 8:

Data Models and its real-world applications



Question 9:


SCHEMAS AND INSTANCES
and their Relationship with Data Independence
SCHEMA (Logical Structure)
• Describes the structure of the database.
• Includes tables, attributes, relationships, constraints, and rules.
• Changes to schema affect how data is organized logically.
Achieved Through
LOGICAL DATA INDEPENDENCE
• Ability to change the conceptual schema without changing external schemas or application programs.
• Insulates external level from changes at conceptual level.
Achieved Through
PHYSICAL DATA INDEPENDENCE
• Ability to change the internal (physical) schema without changing the conceptual schema.
• Insulates conceptual level from changes at internal level.
Schema describes the structure of the instance
INSTANCE (Data)
• Actual data stored in the database at a particular point in time.
• Data changes frequently (insertions, deletions, updates) but structure (schema) remains the same.
EXTERNAL SCHEMAS (Views)
• User specific views of the database.
• Shows only the data required by the user.
External views are based on the schema (logical structure)
INTERNAL SCHEMA (Physical Level)
• Describes the internal storage structure.
• Hidden from users and applications.
Instance is stored in
PHYSICAL STORAGE (Internal / Physical Schema)
• Describes how data is actually stored.
• Includes file organization, indexes, record formats, access paths, etc.
Physical storage is described by the internal schema
KEY RELATIONSHIPS
Descriptive / Structural Relationship
Provides Logical Data Independence
Provides Physical Data Independence
SUMMARY
Schema defines the structure.
Instance represents the data in that structure.
Logical data independence protects views/programs from schema changes.
Physical data independence protects schema from storage changes.

Question 10:

Roles and Responsibilities in a DBMS environment: DBA, End User, Application Developer